FMA4200 Financial Data Analysis: Midterm Exam (Spring, 2025)

Answer the questions in the answer book.

Question:	1	2	3	4	5	6	Total
Points:	10	25	25	20	20	15	115
Score:							

Name: _____ Student ID: _____ Time limit: 2 hours

1. True or False. No justification is needed.
   (a) (2 points) The SPX returns are empirically light-tailed.
   (b) (2 points) ...
   (c) (2 points) ...
   (d) (2 points) ...
   (e) (2 points) ...

   Total for Question 1: 10

2. Answer the following short questions briefly.
   (a) (5 points) Suppose we have a sample of 1000 stock returns: $x_1, x_2, \ldots, x_{1000}$. How can we estimate the skewness of the stock returns?
   (b) (5 points) ...
   (c) (5 points) ...
   (d) (5 points) ...
   (e) (5 points) ...

   Total for Question 2: 25

3. Consider an ARMA$(1, 2)$ model as follows:

$$X_t = bX_{t-1} + \varepsilon_t + \theta\varepsilon_{t-1} + \phi\varepsilon_{t-2},$$

   where $\varepsilon_t \sim N(0, \sigma^2)$ and ε_t is i.i.d.
   (a) (5 points) When is X_t stationary?
   (b) (5 points) When is X_t invertible?
   (c) (10 points) Calculate $\mathbb{E}[X_t]$, $\mathrm{Var}[X_t]$, $\mathrm{Cov}(X_t, X_{t-k})$ and $\mathrm{Corr}(X_t, X_{t-k})$ for all $k = 1, 2, \ldots$?
   (d) (5 points) What is the best prediction of X_{t+1} given the information up to time t? And what is the variance of the prediction error?

   Total for Question 3: 25

4. (20 points) ...

5. (20 points) ...

6. Consider a European call option on a non-dividend paying stock with strike price K and maturity T. The payoff of the option is given by

$$C(S_T, T) = \max(S_T - K, 0),$$

where S_T is the stock price at time T. We assume that under the physical probability measure $\mathbb{P}$, the stock price follows:

$$S_T = S_0 e^{\mu T + \sigma\sqrt{T}Z},$$

where Z is a normal random variable with mean 0 and variance 1, i.e., its density function is given by

$$f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} \quad \text{for all } z \in \mathbb{R}.$$

The risk-free interest rate is r.

(a) (5 points) Consider a pricing kernel

$$M_T = e^{\alpha + \beta Z},$$

where α and β are constants, and let $\frac{d\mathbb{Q}}{d\mathbb{P}} = M_T$. Determine the values of α and β such that $\mathbb{Q}$ is a risk-neutral probability measure, i.e.,

$$\mathbb{E}_{\mathbb{Q}}[S_T] = S_0 e^{rT}, \ \mathbb{E}_{\mathbb{P}}[M_T] = 1.$$

(b) (10 points) What is the distribution of $\ln S_T$ under the risk-neutral probability measure $\mathbb{Q}$?

(c) (10 points (bonus)) Derive the closed-form expression for the price of the call option and propose a method to estimate the parameters given that we observe the market price of the options with different strike prices $K = K_1, K_2, \ldots, K_n$.

Total for Question 6: 15